PrefSR: Direct Preference Optimization for Single-Step Diffusion Image Super-Resolution

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1. Factsheet Information

1.1. Team details

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- Affiliation of the team and/or team members with NTIRE 2026 sponsors (check the workshop website) : No
- User names and entries on the NTIRE 2026 Codabench competitions (development/validation and testing phases): wangwb, 633212
- Best scoring entries of the team during the testing phase: PSNR: 24.58; SSIM: 0.7003; NIQE:3.970; ManIQA:0.6093; MUSIQ:71.16; CLIP-IQA: 0.8402; MANIQA: 0.6092; LPIPS:Unknown; DISTS:Unknown.
- Link to the codes/executables of the solution(s): https://github.com/wwbwang/NTIRE2026_IMAGESR_X4

2. Method

2.1. Method Overview

As illustrated in Fig. 1, our framework adopts a two-module cascaded architecture consisting of a frozen **MambaIRv2** upsampler and a **Stable Diffusion 2.1** UNet equipped with **LoRA** adapters.

Given a low-resolution input x_{LQ}, MambaIRv2 first performs $4\times$ spatial upsampling, yielding a coarse high-resolution estimate $\hat{x}_{\mathrm{SR}}$. This intermediate output provides a *pre-denoised* initialization that suppresses low-frequency degradations and offers a strong structural prior for the subsequent refinement stage. The upsampled image is then encoded into the latent space, where a single-step diffusion denoising process operating at a fixed timestep refines the latent representation to recover realistic high-frequency details. The final latent is decoded to produce the restored high-resolution output $\hat{x}_{\mathrm{HR}}$.

In this framework, the diffusion UNet serves as the primary learnable module, while MambaIRv2 remains frozen throughout training, preserving its pretrained restoration capability and reducing the overall optimization complexity.

2.2. Stage 1: Adversarial Pretraining Stage

2.2.1. Degradation Setting

While Real-ESRGAN [12] establishes a widely adopted high-order degradation pipeline for blind super-resolution, we argue that its default configuration is not universally optimal across all degradation scenarios. When the degradation type of the input low-quality image is partially observable, *e.g.*, images degraded by bicubic downsampling, carefully tailoring the degradation pipeline to match the observed LQ characteristics is critical to achieving optimal restoration quality.

Through systematic analysis of bicubic-degraded images, we observe that sinc-based low-pass filtering and JPEG compression artifacts contribute minimally to this degradation type. Incorporating such components into the training degradation pipeline introduces a domain gap between synthetic training pairs and real inputs, which we empirically find to degrade restoration performance.

Motivated by these observations, we design a two-stage degradation pipeline adapted from Real-ESRGAN. In the

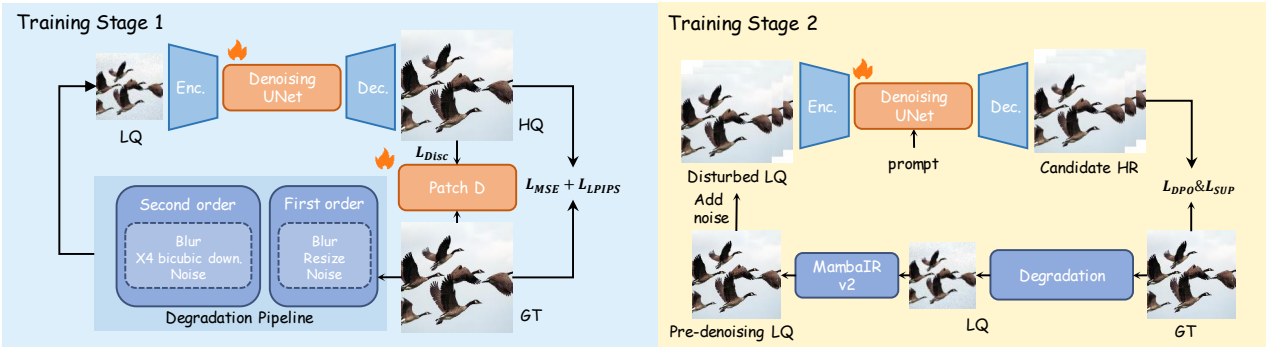

Figure 1. **Overview of the proposed training framework.** As illustrated, our method consists of two training stages: (1) adversarial pretraining stage and (2) preference-based optimization stage.

first stage, we apply blur, resize, and noise degradations; notably, low-pass filter kernels (*i.e.*, sinc kernels) are excluded from the blur kernel pool to avoid introducing ringing artifacts irrelevant to the target domain. In the **second stage**, we apply blur and additive noise, followed by a fixed $\times 4$ bicubic downsampling to produce the final LQ image. This design ensures that the synthesized training data closely reflects the real degradation distribution while retaining sufficient diversity for robust generalization.

2.2.2. Adversarial Single-Step Super-Resolution Model

It is essential for equipping the model with the capacity to generate perceptually plausible high-frequency details. To preserve the generative capability inherited from large-scale pretraining while adapting the model to the super-resolution task, we follow the training strategy of HYPIR [7].

Specifically, we adopt Stable Diffusion 2.1 [10] as the generator backbone, leveraging its strong image synthesis prior. To encourage the generation of realistic textures and sharp structures, we introduce adversarial training with a patch-based discriminator [5].

The overall generator objective is formulated as:

$$\mathcal{L}_G = \lambda_{\text{MSE}}\mathcal{L}_{\text{MSE}} + \lambda_{\text{LPIPS}}\mathcal{L}_{\text{LPIPS}} + \lambda_{\text{GAN}}\mathcal{L}_{\text{GAN}}, \quad (1)$$

where $\mathcal{L}_{\text{MSE}}$ is the pixel-wise mean squared error loss that enforces low-frequency fidelity, $\mathcal{L}_{\text{LPIPS}}$ is the perceptual loss [14] computed in a pretrained feature space to capture perceptual similarity, and $\mathcal{L}_{\text{GAN}}$ is the adversarial loss that drives the generator to produce more visually plausible high-frequency details. The scalar weights λ_{MSE}, λ_{LPIPS}, and λ_{GAN} balance the contribution of each term.

2.3. Stage 2: Preference-Based Optimization

2.3.1. Direct Preference Optimization

While supervised restoration losses are effective for preserving content fidelity, they are often insufficient to fully optimize perceptual quality. To further bias the model toward visually preferred solutions, we introduce **Direct Preference Optimization (DPO)**[9] into the diffusion refinement stage. Inspired by recent DPO-based alignment methods for diffusion models, we cast restoration refinement as a *pairwise preference optimization* problem, where the model is explicitly encouraged to favor perceptually better outputs over worse ones.

Since image restoration is inherently one-to-many, a single low-resolution input may correspond to multiple plausible high-resolution solutions. To expose this ambiguity, we perturb the images with Gaussian noise and generate multiple candidates:

$$c_{\text{img}}^{(k)} = c_{\text{img}} + \sigma\epsilon^{(k)}, \quad \epsilon^{(k)} \sim \mathcal{N}(0, I). \quad (2)$$

Each perturbed image is passed through the one-step diffusion refiner to obtain a candidate output $x^{(k)}$. We then rank all candidates using a perceptual quality score from SNUCV [1] and select the highest-scoring sample as the *chosen* x^+ output and the lowest-scoring one as the *rejected* output x^-.

Based on these preference pairs, we optimize the diffusion refiner with DPO. Instead of using explicit likelihoods, we adopt the negative denoising error as a surrogate log-probability:

$$\log p_\theta^{\text{proxy}} = -\left\|\hat{\epsilon} - \epsilon^*\right\|_2^2, \quad (3)$$

where $\hat{\epsilon}$ is the predicted noise and ϵ^* is the DDPM target noise. For each chosen/rejected pair, we compute

$$\Delta_\pi = \log p_\theta(x^+) - \log p_\theta(x^-), \quad (4)$$

and compare it against a reference policy:

$$\Delta_{\text{ref}} = \log p_{\text{ref}}(x^+) - \log p_{\text{ref}}(x^-). \quad (5)$$

The DPO loss is then defined as

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}\left[\log\sigma\big(\lambda(\Delta_\pi - \Delta_{\text{ref}})\big)\right]. \quad (6)$$

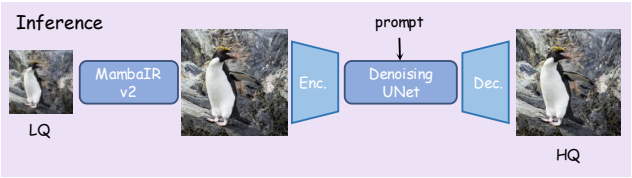

Figure 2. **Overview of the inference procedure of our proposed model.** We first perform a coarse super-resolution of the low-quality (LQ) input using MambaIRv2-Base, and then apply our model to further refine fine-grained details and semantic fidelity.

This objective directly encourages the current policy to prefer perceptually better restoration candidates over worse ones, relative to the reference model.

2.3.2. Overall Objective

Our final training objective combines supervised restoration and preference optimization:

$$\mathcal{L}_G = \mathcal{L}_{\text{sup}} + \lambda_{\text{DPO}} \mathcal{L}_{\text{DPO}}, \tag{7}$$

where the supervision term is defined as:

$$\mathcal{L}_{\text{sup}} = \|x - x_{\text{GT}}\|_1 + \alpha\, \mathcal{L}_{\text{LPIPS}}(x, x_{\text{GT}}) + \beta\, \mathcal{L}_{\text{CLIP}}. \tag{8}$$

Here, $\mathcal{L}_{\text{CLIP}}$ denotes the CLIP-based perceptual loss adopted from SNUCV [1]. In practice, we use an EMA version of the current model as the reference policy to stabilize preference learning. With this design, the supervised objective maintains reconstruction fidelity, while DPO further drives the diffusion refiner toward perceptually preferred solutions.

2.3.3. Training Strategy

During each training iteration, we first perform a deterministic forward pass to compute the supervised loss. We then sample multiple image perturbations to construct candidate outputs and compute the DPO loss. The total loss is backpropagated through the LoRA-adapted diffusion UNet, followed by gradient clipping, optimizer update, and EMA synchronization. The MambaIRv2 upsampler remains frozen throughout training, allowing the diffusion refinement module to focus on perceptual enhancement.

3. Experiments

3.1. Training Data

To support the training of our large-scale image restoration model, we adopt a mixed training set composed of high-quality natural images and facial images. Overall, the training data follows the data construction strategy of SUPIR [13] and HYPIR [7], using approximately 20 million high-quality training samples as the main corpus, with each sample paired with a textual description to provide additional semantic guidance. Compared with existing image restoration datasets, this training set offers a better balance between scale and visual quality, covering a wider range of scenes, textures, and semantic content.

In our implementation, the training data is organized in the WebDataset format and consists of two high-quality natural image subsets derived from Unsplash, together with a non-aligned high-resolution facial image subset from FFHQ. Such a data composition enables the model to achieve strong restoration performance on general natural scenes while also improving perceptual quality in facial regions.

3.2. Implementation Details

Stage 1. We train the adversarial single-step super-resolution model using the AdamW [6] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$. The generator and discriminator are optimized with learning rates of 1×10^{-4} and 1×10^{-5}, respectively. Ground-truth images are randomly cropped into patches of size 512×512 during training. The model is trained for 18k iterations with a batch size of 6 and an EMA decay of 0.999. LoRA adapters are configured with rank $r = 256$. The loss weights are set to $\lambda_{\text{MSE}} = 1$, $\lambda_{\text{LPIPS}} = 5$, and $\lambda_{\text{GAN}} = 0.5$. During inference on high-resolution images, we adopt a sliding window strategy with a patch size of 512×512 and a stride of 256 to ensure seamless tiling without boundary artifacts.

Stage 2. We further enable DPO with $K = 3$ candidates by injecting Gaussian noise ($\sigma = 0.02$) into the latent input and using CLIPIQA as the preference reward; the DPO update is weighted by $\lambda_{\text{DPO}} = 0.5$ with $\beta = 0.1$, and the EMA model (decay 0.999) serves as the reference policy when available.

3.3. Experimental results

We validate our method on the DIV2K validation set (0801-0900) under the x4 restoration setting. Our validation pipeline consists of two stages. We first apply MambaIRv2 to generate a 4× upsampled result from the low-quality input, and then use an SD2.1-based HYPIR model to perform a 1× refinement on the upsampled image. The second stage is conditioned on an LLaVA-generated image description and is performed in a tiled manner with a patch size of 512 and a stride of 256. After decoding, we further apply a wavelet-based reconstruction step to enhance perceptual details while preserving stable low-frequency structures.

3.4. Evaluation Metrics

We use both full-reference and no-reference metrics for evaluation. Specifically, PSNR and SSIM are used to measure reconstruction fidelity, while LPIPS and DISTS are used to assess perceptual similarity. We further adopt CLIP-IQA, MUSIQ, MANIQA, and NIQE for no-reference quality evaluation. Following the evaluation protocol provided

by the challenge organizers, we also report a weighted perception score as the final overall metric.

4. Other details

- We plan to submit a solution description paper to the NTIRE 2026 workshop.

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